

Real-Time Smear Diagnostics and Electric-Field Suppression of SiO_x/CN_x Accumulation at the HAMR Head-Disk Interface

Bernhard. Knigge*, Daniel Maksuta, Yuichi Aoki, Wangchen Zhao

Western Digital Corporation, 5601 Great Oaks Blvd, San Jose, CA 95119, *bernhard.knigge@wdc.com

Abstract—Heat-assisted magnetic recording (HAMR) drives material accumulation onto the recording head — SiO_x and CN_x and disk lubricant leading to recording degradation and lifetime reduction of the near-field transducer (NFT). Interface voltage control (IVC), a DC bias between slider and disk, modulates the electric field across the sub-1 nm head-disk gap and controls the smear pathways. The embedded contact sensor (ECS) and NFT thermal sensor (NTS) resolve distinct smear signatures: a localized NTS cooling dip during writer approach uniquely identifies thermal bridging by smear, while IVC Joule heating leading to temperature increases - is eliminated when the contact potential is offset. Long-term lifetime trials confirm that when the contact potential is offset - field-driven SiO_x accumulation and CN_x contamination is reduced, reducing ECS and NTS heating and lowering smear-driven degradation signals.

Index Terms—CN_x smear, SiO_x buildup, HAMR, head-disk interface, interface voltage control, thermistor, lifetime.

I. Introduction

HAMR concentrates optical energy via a plasmonic near-field transducer (NFT)—typically a gold nanostructure—to transiently heat FePt recording media above its Curie point [1]. Operating conditions at the head-disk interface (HDI) are extreme: NFT temperatures near 700 K, sub-1 nm gaps, and an intrinsic contact potential difference (CPD) of 0.2–0.6 V that generates electric fields of 10⁸–10⁹ V/m. Two dominant smear species change recording performance over head lifetime: (i) SiO_x accumulates on the NFT, shifting its plasmonic resonance [2]; (ii) CN_x and lube fragments from the media overcoat introduce optically absorbing contamination near the NFT tip [3,11]. Interface voltage control (IVC)—a controlled DC bias between slider body and disk—affects HDI electric field and offers a direct lever for managing both pathways. [4]. The optimal IVC voltage (OIV) is the bias that exactly cancels the CPD, setting the net HDI electric field to zero and eliminating field-driven ionic transport. An external AC electric field has also been shown to reduce smear by trapping desorbed ions in the gap before they reach the slider surface [13]. Embedded thermal sensors (ECS and NTS) enable real-time in-situ monitoring of smear state, complementing laser current changes as a plasmonic health indicator.

II. Current Understanding of SiO_x Growth

Silicon dioxide (SiO₂) formation on silicon surfaces is well-understood through established kinetic frameworks directly applicable to smear buildup in heat-assisted magnetic recording (HAMR). The Deal–Grove model quantifies SiO₂ growth via a linear-parabolic law in which oxidant partial pressure (O₂ or H₂O) and temperature are primary rate drivers; wet conditions accelerate growth 2–10× over dry O₂, and elevated pressure increases the parabolic rate constant proportionally [5,6]. Applied voltage controls SiO₂ growth rate, structure, and stoichiometry: anodically grown oxides shift toward silicon-rich, sub-stoichiometric compositions under varying deposition conditions, directly affecting refractive index and optical properties [7]. The Cabrera–Mott model establishes that ionic transport — and therefore oxide growth — occurs only at the positive (anodic) surface; built-in electric field magnitude governs the rate [8]. AFM local anodic oxidation (LAO) confirms strict polarity control: positive bias drives OH[−] from adsorbed water into silicon forming SiO₂; reversed polarity suppresses oxidation completely [9,10,12]. In HAMR, the

recording head acts as the electrode in a nanometer-scale electrochemical cell. Positive head bias creates anodic conditions, driving OH[−] from drive humidity toward the head and accelerating SiO₂ smear accumulation; negative bias suppresses this. Interface voltage control - alongside drive O₂ concentration, humidity, and air-bearing pressure - is therefore a handle on SiO₂ quantity and quality at the head–disk interface.

III. ECS/NTS Smear Thermal Diagnostics

The embedded contact sensor (ECS, ~2 μm downstream of the read element) and NFT thermal sensor (NTS, ~2 μm downstream of the NFT) are resistive thermistors with positive TCR. Their 5–7 μm spatial separation enables discrimination of four distinct HDI thermal processes (Table I).

Process	ECS	NTS	Key Discriminant
Joule heating (V _{net} ≠ 0)	↑ equal	↑ equal	Eliminated at OIV; laser-independent
Smear cooling	↓ dip	↓ dip	Depends on TFC push
Frictional heating	↑ step	↑ step	At strong TFC push or large smear interference
Lube bridging	↓ cooling	weak	ECS dominant; lube-type dependent

TABLE I. ECS/NTS thermal signature discriminants. The (dECS–dNTS) differential isolates smear cooling from Joule and lube contributions.

IV. Results

A. Spatial Localization of Smear Cooling

Figs. 1a and 1b show ECS and NTS responses during write heater dithering (TFC2). The ECS (Fig. 1a) oscillates symmetrically with TFC2 power without asymmetry between approach and retract half-cycles—the reader zone lies outside the smear footprint.

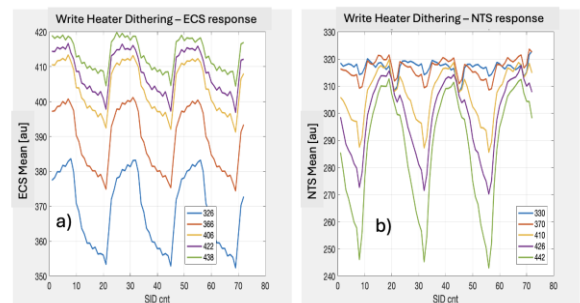


Fig. 1a. ECS thermistor (a) and NTS thermistor (b) temperature response to write heater dithering (TFC2). The oscillation is symmetric for ECS and asymmetric for NTS.

In Fig. 1 the different colors indicate different heater powers. As we approach the disk the amplitude is reduced due to a stiffer airbearing and proximity cooling effects. The rise time and fall times are fairly similar indicating the reader is not contacting smear. The NTS thermistor (Fig. 1b) shows a sharp, asymmetric cooling dip on each approach half-cycle. This NTS-only, approach-selective signature is evidence of smear thermal bridging at the writer: no alternative mechanism—Joule heating, lube bridging, or TFC self-heating—produces this pattern.

B. Spacing-Dependent Smear Accumulation

Fig. 2 shows NTS and acoustic emission touchdown curves before and after aging at five different spacings / backoff power levels (10–50 mW). We observe that the NTS cooling signature is earlier, after the head was subjected to higher flyheights. Clearly, the gap between slider and disk is always filled with smear – and smear height is proportional TFC2 backoff. Normalizing by the HMS actuation efficiency nm/mW gives smear thickness $\delta_{\text{smear}} \propto \text{backoff spacing}$ —establishing the HMS setpoint as a direct lever for smear management.

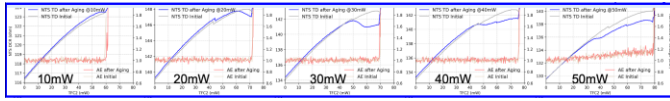


Fig. 2. NTS touchdown curves before/after aging at five backoff spacings. NTS cooling dip onset (blue) shifts earlier with increasing spacing, quantifying smear protrusion height as a function of HMS.

C. Long-Term IVC Effects on SiO_x and CN_x

Fig. 3 shows median alive-head traces (~800 hr) for four conditions: each line is the median of ~40 heads tested in drive. dry no-IVC, 3 % RH no-IVC, 3 % RH IVC (–400 mV), and 5 % RH IVC. Four panels reveal clear HDI state separation:

- FH (top-left): dry no-IVC shows the fastest head-media spacing rise mainly driven by large CN_x/lube accumulation. Higher humidity reduces dFH changes and IVC has also a small benefit in reducing FH changes.
 - Laser power (top-right): All conditions show a steady laserpower drop (dLP) – dry seems to have the least drop and hence the poorest optical coupling. 3% RH has the most drop and 3% RH with IVC has a significant impact on LP changes – likely due to reduced SiO₂ formation at negative IVC.
 - NTS heating (bottom-left): dry no-IVC produces the strongest NTS rise from frictional smear contact at very large interference. At 5% RH with negative IVC we barely see a frictional heat signature.
 - ECS cooling (bottom-right, red arrow): ECS resistance is dropping for all interfaces indicating progressive lube/smear cooling. However, the most drop is seen at negative IVC as we operate at OIV point and hence have eliminated Joule heating.
- It is interesting that while IVC affects both the CN_x/lube smear which is key driver for FH changes – it's effect is much bigger on the SiO_x growth as indicated by a significant laser current change.

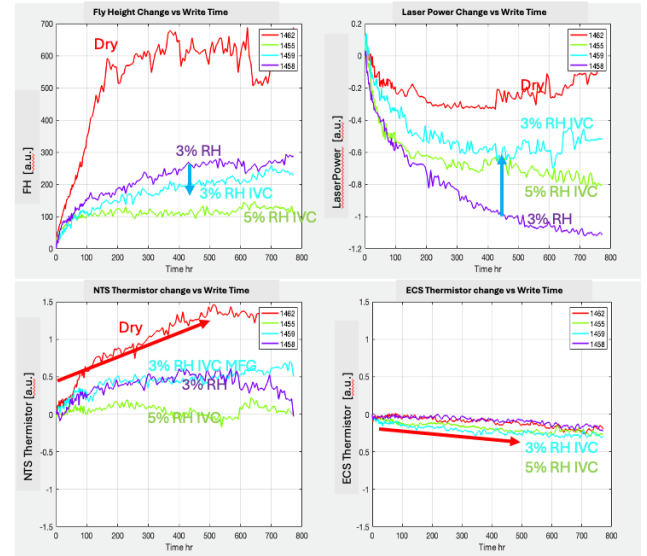


Fig. 3. Median of alive heads (~800 hr write testing) for four IVC/humidity conditions. Four panels: FH (top-left), laser power (top-right), NTS resistance (bottom-left), ECS resistance (bottom-right).

V. Conclusion

ECS and NTS sensors provide spatially resolved, real-time diagnostics for SiO_x and CN_x smear in HAMR drives. The NTS cooling dip during TFC2 approach uniquely identifies thermal bridging by smear at the writer, while IVC Joule heating—eliminated at OIV—appears equally at both sensors and serves as a calibration reference. Smear protrusion height scales monotonically with operating head-media spacing, making HMS setpoint a primary smear management lever. IVC affects both lube/CN_x and SiO_x growth rates – as indicated by dFH and dLP.

I. REFERENCES

- [1] W. A. Challener et al., “Heat-assisted magnetic recording by a near-field transducer with efficient optical energy transfer,” *Nat. Photon.*, vol. 3, pp. 220–224, 2009.
- [2] B. C. Stipe et al., “Magnetic recording at 1.5 Pb m⁻² on a near-field transducer,” *Nat. Photon.*, vol. 4, pp. 484–488, 2010.
- [3] B. Marchon et al., “The physics of disk drive tribology,” *Annu. Rev. Mater. Res.*, vol. 43, pp. 19–45, 2013.
- [4] B. Knigge et al., “Influence of contact potential on slider–disk spacing,” *IEEE Trans. Magn.*, vol. 40, no. 4, pp. 3165–, 2004.
- [5] B. E. Deal and A. S. Grove, “General relationship for the thermal oxidation of silicon,” *J. Appl. Phys.*, vol. 36, pp. 3770–3778, 1965.
- [6] H. Kageshima et al., “Kinetics of thermal oxidation of silicon,” *Appl. Phys. Express*, vol. 18, p. 120101, 2025.
- [7] A. Ashok and P. Pal, “Growth and etch rate study of low temperature anodic silicon dioxide thin films,” *J. Nanomater.*, vol. 2014, p. 670403, 2014.
- [8] N. Cabrera and N. F. Mott, “Theory of the oxidation of metals,” *Rep. Prog. Phys.*, vol. 12, pp. 163–184, 1949.
- [9] J. A. Dagata et al., “Modification of hydrogen-passivated silicon by a scanning tunneling microscope operating in air,” *Appl. Phys. Lett.*, vol. 56, pp. 2001–2003, 1990.
- [10] J. A. Dagata, “Device fabrication by scanned probe oxidation,” *Science*, vol. 270, pp. 1625–1626, 1995.
- [11] B. Marchon et al., “Lubricant design attributes for HAMR hard disk drives,” *IEEE Trans. Magn.*, vol. 45, no. 10, pp. 3670–3673, 2009.
- [12] R. Garcia et al., “Advanced scanning probe lithography,” *Nat. Nanotechnol.*, vol. 9, pp. 577–587, 2014.
- [13] H. Tani et al., “Head Smear Reduction in Head-Disk Interface of HAMR Using External AC Electric Field,” *IEEE Trans. Magn.*, vol. 61, no. 4, art. no. 3200205, 2025. doi: 10.1109/TMAG.2024.3452784.